

Experiment no : 10

Aim : To study star Topology and Ring Topology

Theory:

Software: Cisco Packet Tracer

THEORY:

Star topology is a network configuration where all devices are connected to a central device, called a hub or switch. In this setup, devices are connected radially to the central point, and data is transmitted from one device to another through the central hub. This topology offers advantages such as ease of installation and maintenance, fault tolerance, and scalability, making it suitable for large networks, including local area networks (LANs) and wireless networks. However, it also has its drawbacks, including a central point of failure, dependence on the hub, and higher costs.

FEATURES:

1. Central connection point: All devices connect to a central hub or switch.
2. Devices connected radially: Devices are connected to the central point using separate cables.
3. Data transmitted through the hub: Data is transmitted from one device to another through the central hub.

ADVANTAGES:

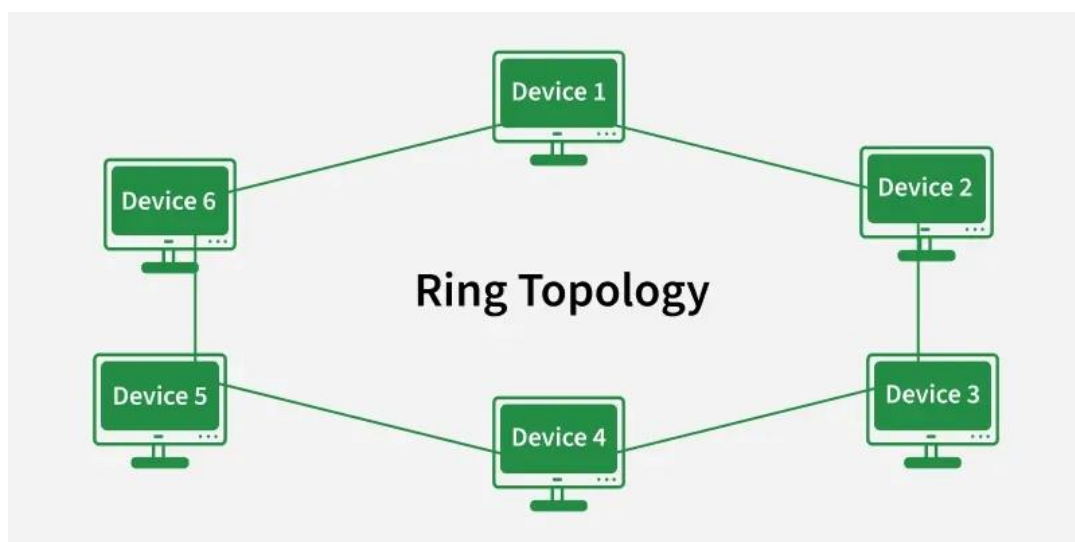
1. Easy to install and maintain: Star topology is easy to install and maintain.
2. Fault tolerance: A faulty device can be easily disconnected without affecting the rest of the network.
3. Scalability: Star topology is suitable for large networks.

DISADVANTAGES:

1. Central point of failure: If the hub or switch fails, the entire network fails.
2. Dependence on the hub: All devices rely on the central hub to function.
3. Higher cost: Star topology requires more cables and a central hub.

In a Ring Topology, it forms a ring connecting devices with exactly two neighboring devices. A number of repeaters are used for Ring topology with a large number of nodes, because if someone wants to send some data to the last node in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node. Hence to prevent data loss repeaters are used in the network.

In-Ring Topology, the Token Ring Passing protocol is used by the workstations to transmit the data where, Token passing is a network access method in which a token is passed from one node to another node & Token is a frame that circulates around the network



Operations of Ring Topology

- One station is known as a monitor station which takes all the responsibility for performing the operations.
- To transmit the data, the station has to hold the token. After the transmission is done, the token is to be released for other stations to use.
- When no station is transmitting the data, then the token will circulate in the ring.

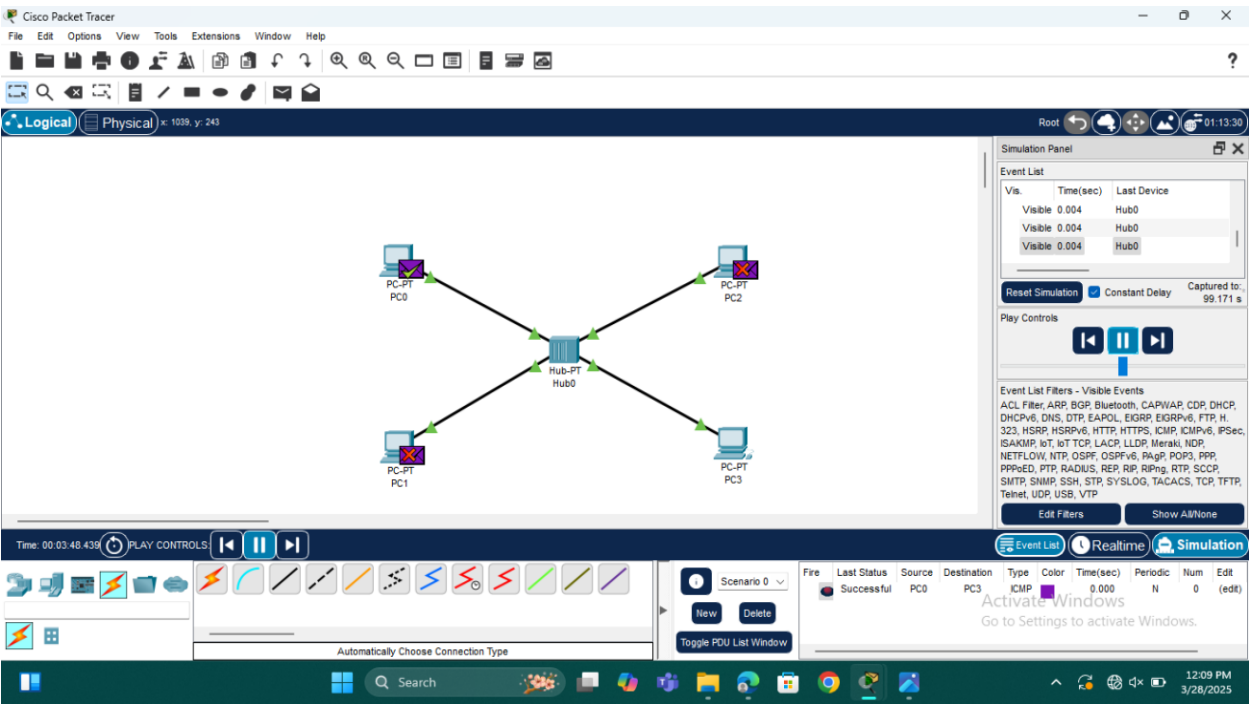
Advantages of Ring Topology

- The data transmission is high-speed.
- The possibility of collision is minimum in this type of topology.
- Cheap to install and expand.

Disadvantages of Ring Topology

- The failure of a single node in the network can cause the entire network to fail.
- Troubleshooting is difficult in this topology.
- Less secure.

OUTPUT:



Conclusion: